

Cooling a Quasi-Direct-Drive Joint Motor from Inside the Copper: A Simulation-Only Design Study

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<https://github.com/punkfab/robot-actuators>

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Abstract

Quasi-direct-drive (QDD) robot joints trade gear ratio for transparency, and their continuous torque is set by how much winding heat the housing can shed. We test one hypothesis: that drive electronics are now efficient enough at high current that heat is the remaining limit, so a joint motor should be cooled directly, run hard, and geared less. The design studied is a $\varnothing 120$ mm, 24-slot, 22-pole motor whose coils are hollow rectangular copper carrying dielectric oil in the bore, inside an outer jacket fed with the coldest coolant. It is evaluated with a first-order loss and hydraulic model, a nonlinear two-dimensional magnetostatic field solution that agrees with FEMM within 0.7%, a thermal field solution of the jacket-only alternative, and a parametric CAD assembly with interference checks, all driven by one design record. *Nothing has been built or measured.* In simulation the motor gives 16.6 N m continuously for 850 W of heat with the copper at 126 °C, and 21.3 N m at the limit of a 3 L/min pump, against about 5 N m for the same iron cooled by air. Three results ran against the hypothesis. The saturation limit first assumed was too pessimistic, by a quarter to a third of the torque. Trading copper for iron at fixed size does not raise torque at a given heat: eleven geometry variants land within about 10 % of each other. And a much simpler motor, with solid conductors and only the jacket, reaches about 75 % of the hollow design's torque, so it is the one to build first. A 3:1 reduction then matches a commercial 9:1 actuator's continuous torque with a ninth of the reflected inertia, and spends about five times the power holding the same load. The models and CAD were written by an AI coding agent under the author's direction; we describe that workflow and the modelling errors that cross-checks between the models exposed. No new cooling method is claimed. The contributions are the design, its comparison with the jacket-only motor, and the open code.

1 Introduction

The proprioceptive, or quasi-direct-drive, actuator pairs a large-diameter torque motor with a single low-ratio gear stage, so that the joint is backdrivable, its reflected inertia is small, and motor current is a usable measure of output torque [1–3]. The price is that torque comes from current instead of from gearing. Copper loss rises with the square of torque, and the continuous rating of such a joint is a thermal figure: what the winding can lose through the slot insulation, the iron, the housing and whatever the housing is bolted to.

This study tests a design hypothesis: high-current solid-state drives are now efficient enough that heat, not the drive, limits a joint motor, so the conductors should be cooled directly with liquid, the hot machine isolated from its surroundings by an outer jacket, the copper run at high current density and temperature, and the gear ratio reduced.

The contributions are:

1. a hollow-conductor, oil-cooled motor design in the envelope of a commercial QDD actuator, evaluated by loss, hydraulic and nonlinear field models and drawn as a checked CAD assembly;
2. a comparison, on the same iron, with a jacket-only motor using solid conductors, and the consequences of both for gear ratio and holding power;
3. the open models and solver, and a description of the agent-assisted workflow that produced them, including the errors found by cross-checking (Section 6).

Scope. The study is simulation only: nothing has been built or measured. Cooling a winding by passing coolant through hollow conductors is decades old in large machines and has been demonstrated in small ones (Section 2); no novelty is claimed for the method. Every thermal coefficient is a handbook value or a stated assumption, and Section 7 lists what the models leave out.

2 Related work

Quasi-direct-drive actuators. The design principles are those of the MIT Cheetah: a high-torque-density motor, a low gear ratio, and torque control through motor current [1, 2], later made low-cost and modular by Katz [3]. Seok et al. [2] report that 76 % of their robot’s energy in trotting is motor heat, which is the loss this study is about.

Liquid cooling of robot actuators. Cooling a robot’s motors with liquid is established. Urata et al. [4] combined forced liquid cooling with estimation of the motor’s internal temperature in a high-power humanoid leg. Paine and Sentis [5] retrofitted liquid cooling to an off-the-shelf motor and measured 2.58 times the continuous torque, and Kim et al. [6] and Zhu et al. [7] built and modelled liquid-cooled actuators. Mazumdar et al. [8] report 50 % more heat transfer from a redesigned motor housing, and 107 % more with pumped water. Kozuki et al. [9] used a humanoid’s skeleton to release heat by latent heat, and Li et al. [10] cool humanoid joint motors by evaporation from phase-change hydrogels. Closest to the winding, Zhu et al. [11] immersed the coils of a legged-robot motor directly in coolant, built and tested it, and report a thermal resistance 3.7 times better than liquid cooling and 10 times better than forced air. That last work is the nearest to ours: it already brings coolant into contact with the winding of a legged-robot motor.

Coolant inside the conductor. Hollow stator conductors carrying water are long-standing practice in large turbo-generators [12, 13]. In smaller machines, Lindh et al. [14, 15] cooled the tooth coils of an axial-flux traction machine with oil or water in a stainless tube wrapped in Litz wire and compared it with water jackets; Schiefer and Doppelbauer [16] placed water channels in the unused slot space of a concentrated winding; Wohlers et al. [17] studied direct liquid

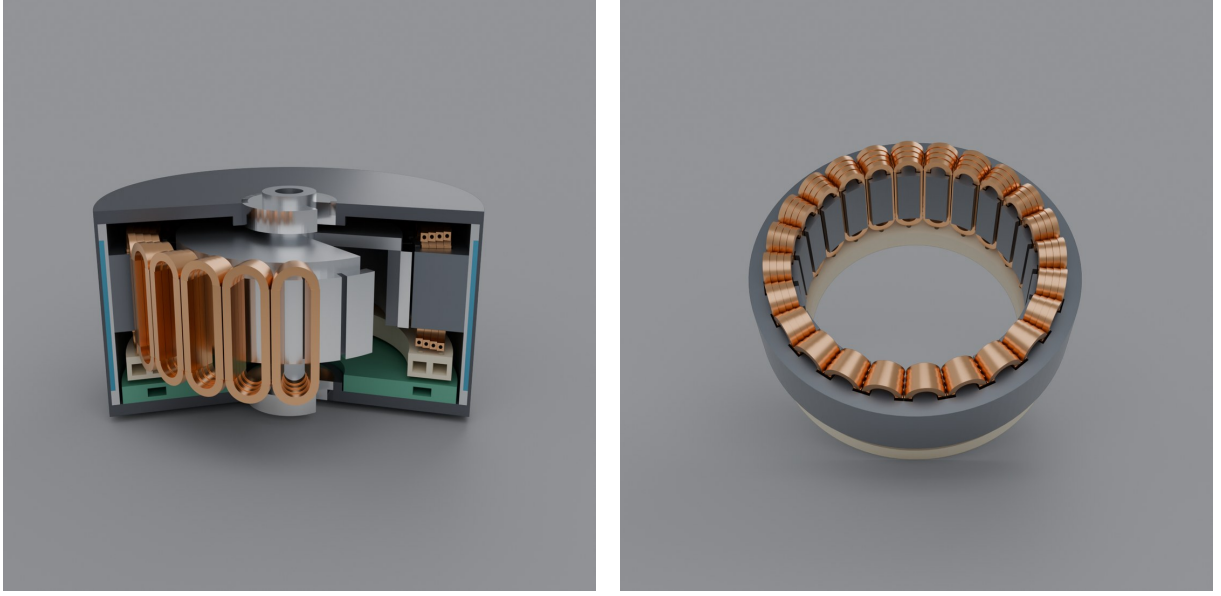


Figure 1: The shallow-slot design, rendered from the CAD assembly. Left: cutaway with the hollow conductors in section, the magnets, the coolant (blue) in the jacket channel, the manifold ring and the drive plate. Right: the wound stator with its 24 single-column coils.

cooling of tooth-coil windings; and Wu and EL-Refaie [18], Wu et al. [19] designed additively manufactured hollow conductors for aircraft machines. Geelen et al. [20] is a simulation study of water-cooled hollow conductors in coreless linear motors. Reviews of machine cooling are given by Gai et al. [21] and Yang et al. [22], and the sizing relations we use are standard [23].

Where this study sits. We found no published robot joint motor with coolant inside the winding conductor. Our search covered the indexed English-language literature and did not cover patents or work in other languages, so we do not claim there is none. What this study adds is narrow: the hollow-conductor scheme sized for a QDD joint envelope, the comparison with the jacket-only motor on the same iron, and the consequence for gear ratio.

3 Concept

The motor (Figure 1) has these features.

- **Hollow copper conductors.** Coolant flows in the bore of every turn, so heat leaves from inside the copper instead of crossing insulation, iron and a housing.
- **Few turns of large conductor.** A low-voltage, high-current machine, with one coil per tooth (24 slots, 22 poles).
- **Dielectric oil.** It can touch live copper, so the coils are electrically in series and hydraulically in parallel with no insulating breaks.
- **The drive on the motor,** with its cold plate on the same loop.

- **An outer jacket fed first.** The flow order is radiator → drive cold plate → outer jacket → conductor bores → radiator. The coldest fluid lines the outside of the machine before it reaches the heat source, in the way a regeneratively cooled rocket nozzle is lined by its propellant [24].
- **Stator outside,** pressed into the jacket, with the rotor inside and a $\varnothing 64$ mm free bore.

The envelope is that of the RobStride RS04 [25], a $\varnothing 120$ mm actuator with a 9:1 reduction rated at 35 N m continuous (on a 220×200 mm heat sink) and 120 N m peak, which serves as the reference throughout.

4 Models

All four models read one frozen design record (`design.Design`), so the field solver, the loss model and the CAD cannot disagree about a dimension.

4.1 Sizing: losses, hydraulics, drive

Unsaturated torque is

$$T = 3 k_w N_{ph} B_1 r_g L \hat{I}, \quad (1)$$

with winding factor $k_w = 0.95$, N_{ph} series turns per phase, B_1 the peak of the air-gap field's working harmonic, r_g the gap radius, L the stack length and $\hat{I}$ the peak phase current. Copper loss is $P_{cu} = \rho(\theta) J^2 V_{cu}$ with $\rho(\theta) = \rho_{20} [1 + \alpha(\theta - 20)]$, $\alpha = 0.00393 \text{ K}^{-1}$, evaluated at the mean copper temperature, which the model solves for.

Each coil is one hydraulic path. For a pressure Δp across it the bore velocity v follows from $\Delta p = f(\ell/d) \rho v^2/2$, with $f = 64/Re$ in laminar flow blended to the Blasius form above $Re = 2300$, and the film coefficient from $Nu = 4.36$ blended to the Dittus–Boelter correlation [26]. The hottest copper is at the coil outlet:

$$\theta_{max} = \theta_{in} + \frac{P_{drv} + P_{fe}}{\dot{m} c_p} + \frac{P_{cu}}{\dot{m} c_p} + \frac{P_{cu}/N_s}{h \pi d \ell}, \quad (2)$$

where the second term is the rise across the drive plate and jacket, which the coolant passes first. Drive loss is conduction in the switches and leads plus switching, $P_{drv} = 3I^2(R_{ds} + R_{lead}) + 3(0.9V_{bus}) I t_{sw} f_{sw}$, with $R_{ds} = 0.5 \text{ m}\Omega$ per switch position, $R_{lead} = 0.2 \text{ m}\Omega$, 48 V, 40 kHz and 20 ns, all assumed. Pump power is $\Delta p Q/\eta$ with $\eta = 0.4$.

Two choices in this model matter to the results. The pump pressure at each operating point is the one that gives the *least total heat* (more flow costs pump power but cools the copper and lowers its resistance), and total flow is capped at 3 L/min, our assumption for what a joint-sized pump and hoses carry. The continuous limit is the current at which the hottest copper reaches 200 °C under that cap. The oil is a polyalphaolefin-class dielectric with handbook properties (at 66 °C: 785 kg/m³, 2190 J/kg/K, 0.138 W/m/K, 2.5 mPa s), entering at 60 °C.

4.2 Field solution

Equation (1) needs B_1 and says nothing about saturation, so the sizing model first carried an assumed air-gap field and an assumed saturation knee (an air-gap shear of 90 kPa). To replace

both we wrote a two-dimensional magnetostatic solver for the cross-section. It solves

$$\nabla \cdot (\nu(|\mathbf{B}|^2) \nabla A_z) = -J_z - \nabla \times (\nu \mathbf{B}_r) \cdot \hat{z} \quad (3)$$

for the axial vector potential on linear triangles generated by Gmsh [27], with $A_z = 0$ on the outer boundary. The iron's reluctivity follows Brauer's fit [28], $\nu = k_1 e^{k_2 B^2} + k_3$ with $(k_1, k_2, k_3) = (3.8, 2.17, 396.2)$, a generic non-oriented electrical steel, continued at the slope of free space beyond the flux density where the fit becomes that steep (about 2.07 T). Newton iteration with backtracking uses the element tangent

$$K_e = \Delta_e \nu G^\top G + 2\Delta_e \frac{d\nu}{dB^2} (G^\top \nabla A)(G^\top \nabla A)^\top, \quad (4)$$

where G holds the shape-function gradients and Δ_e is the element area. Magnets have $B_r = 1.17$ T, $\mu_r = 1.05$ and cover 85 % of the pole pitch. Torque is Arkkio's integral over the air-gap elements [29], $T = \frac{L}{\mu_0(r_o - r_i)} \int r B_r B_\theta dS$. The current angle is chosen by a sweep at each end of the current range.

Three checks on the solver, all on the final design:

- *Mesh refinement.* At 32.5 A/mm², torque is 16.56, 16.58 and 16.55 N m on 17 986, 30 764 and 76 023 triangles: a spread of 0.2 %. The middle mesh is used.
- *Closed form.* At 3 A/mm², well below saturation, the solver gives 1.611 N m. Equation (1) with the solver's own no-load $B_1 = 1.053$ T gives 1.644 N m, 2 % higher.
- *An independent solver.* The same cross-section was drawn in FEMM 4.2 [30] from the same design record, with the same B–H points, winding and current angle. The mesh (131 979 triangles), the nonlinear solver and the torque method (weighted stress tensor) are FEMM's. Table 1 gives the comparison: the no-load air-gap field agrees to four figures and torque within 0.7 % from 3 A/mm² to 100 A/mm². Four other current angles at 32.5 A/mm² agree within 0.3 %.

	this solver	FEMM	difference
no-load air-gap field B_1	1.0529 T	1.0529 T	0.01 %
torque at 3 A/mm ²	1.611 N m	1.621 N m	−0.6 %
14 A/mm ²	7.444 N m	7.450 N m	−0.1 %
26 A/mm ²	13.495 N m	13.486 N m	0.1 %
32.5 A/mm ²	16.578 N m	16.553 N m	0.2 %
43 A/mm ²	21.138 N m	21.072 N m	0.3 %
67 A/mm ²	28.956 N m	28.779 N m	0.6 %
100 A/mm ²	34.494 N m	34.269 N m	0.7 %

Table 1: The field solver against FEMM 4.2 on the shallow-slot design: same geometry and material data, independent mesh, solver and torque method.

The FEMM comparison shows that the solver solves this two-dimensional problem correctly into deep saturation. It does not test what the two share: the two-dimensional assumption, the B–H data and the single rotor position (Section 7). Neither has been compared with a measured machine.

4.3 Jacket-only thermal path

For the alternative with solid conductors, all copper heat crosses, per tooth: conductor insulation and slot liner, tooth, yoke, the press fit to the jacket, and the coolant film. We solve this as a two-dimensional heat-flow field problem in FEMM over one slot pitch: tooth and yoke iron, the liner, the eight insulated conductors with a uniform heat source, the voids between them, and the slot opening. The stator’s outer surface is a convection boundary representing the press fit and coolant film in series; the bore and the tooth centre-lines are adiabatic. The coefficients are a liner and insulation conductivity of 0.2 W/m/K (unpotted, voids filled with air), in-plane lamination conductivity 25 W/m/K, a fit conductance of 2000 W/m²/K and a film coefficient of 1500 W/m²/K. All four are assumed. End-turn heat is taken to conduct along the copper into the slot, so the whole copper loss crosses this section. The result is unchanged for element sizes from 0.5 mm to 0.1 mm.

A lumped model of the same path, five resistances in series, was built first. Its resistance is 12 % higher than the field solution’s (210 against 187 K per W/mm of stack per slot), because it omits heat leaving through the slot back and spreading in the tooth. The field values are used below.

4.4 CAD as a check

A build123d [31] script generates thirteen part types from the same design record: stator core, 24 coils, a two-piece jacket with its annular channel, rotor yoke, 22 magnets, rotor carrier, a manifold ring with supply and return plenums, drive cold plate, two end caps, bearings and shaft. It exports a STEP assembly and computes the shared volume of thirteen part pairs. Its purpose here is to test whether the conductor the loss model assumes can be placed in the slot the field model assumes. Each turn is drawn as a closed racetrack; crossovers, leads, fasteners, seals and bus bars are not drawn.

5 Results

5.1 The assumed saturation limit was wrong

On the first geometry (wedge slots 14 mm deep, half the slot pitch), the field solution gives more torque than the assumed knee at every current (Table 2). The no-load air-gap field is 1.04 T against the 0.83 T assumed, and saturation is far more gradual: torque per amp is still 92 % of its unsaturated value at 26 A/mm², where the assumed knee had the motor close to its ceiling.

current density	field solution	torque per amp	assumed knee	total heat
14 A/mm ²	8.5 N m	98 %	6.8 N m	188 W
26 A/mm ²	14.8 N m	92 %	10.7 N m	674 W
33 A/mm ²	17.5 N m	86 %	12.2 N m	1135 W
42 A/mm ²	20.2 N m	77 %	13.5 N m	2.0 kW
100 A/mm ²	26.4 N m	43 %	16.6 N m	beyond the pump

Table 2: First geometry: the field solution against the saturation knee the sizing model had assumed.

In that geometry torque per amp is down 20 % at 39 A/mm² (19.3 N m) and the 3 L/min pump is exhausted at 49 A/mm² (21.4 N m, about 3 kW). Heat and saturation arrive at nearly the same current. The premise that heat is *the* limit is therefore half right for this iron: remove the heat and the iron is next, immediately.

5.2 More iron in the same size does not help

The natural response is to widen the teeth. Table 3 holds the envelope at $\varnothing 120$ mm with a 25 mm stack and trades copper for iron.

variant	torque (N m) at total heat			cooling limit	K_t down 20 % at
	300 W	850 W	1500 W		
baseline: slots 50% of pitch, 14 mm deep	10.5	15.9	18.8	21.4	39 A/mm ²
slots 40%	9.7	15.4	18.8	22.3	57 A/mm ²
slots 30%	8.5	13.7	17.0	20.6	79 A/mm ²
slots 10 mm deep (larger rotor)	10.5	16.7	20.7	25.0	65 A/mm ²
slots 40%, 10 mm deep	9.6	15.5	19.6	24.5	98 A/mm ²
slots 30%, 7 mm deep	7.7	12.6	16.1	18.8	over 140 A/mm ²
magnets 5 mm	10.7	16.2	19.2	22.0	40 A/mm ²
cobalt-iron laminations	10.8	16.8	20.2	23.5	46 A/mm ²
outrunner	11.8	17.6	20.9	24.0	36 A/mm ²

Table 3: Copper traded for iron at fixed envelope, idealised conductor fill. The cooling-limit column is torque in N m; K_t is torque per amp. Nine of the eleven variants run are shown; the other two (a thicker yoke, and an outrunner with narrow shallow slots) fall inside the same range.

Wider teeth do push saturation out, a long way. They also remove copper, so the same torque needs a higher current density and makes more heat. The two effects nearly cancel: at 850 W every variant but the most iron-rich is within about 10 % of the others. The one clear gain is a shallower slot, which lets the rotor grow: 5 % at 850 W and 17 % at the cooling limit. An outrunner is worth about 11 %, not the factor of 1.5 the first-order model had suggested, and cobalt-iron about 6 %.

At this size the motor makes about 16 N m for 850 W whatever the split between copper and iron. More torque at the same heat means more stack length or more diameter: the trade is size for torque, not size for iron.

5.3 The shallow-slot design

Table 4 and Figures 2 and 3 give the design that follows from the variant study, now with a specific conductor in parallel-sided slots. Here cooling is the limit again: the pump runs out at 43 A/mm² with torque per amp still at 92 %. The flow in the bores is laminar, so the film coefficient is fixed by the bore diameter and the limit is set by the coolant's temperature rise, which is why the flow cap matters more than the pump pressure. The jacket does what it was drawn for: the skin sits within about a degree of the coolant inlet at every operating point. The same iron wound conventionally and limited to 88 W of copper loss gives about 5 N m.

Among six conductor layouts compared, round capillary tube (an off-the-shelf part) fills only 39 % of the slot and gives 12.8 N m at 850 W, a quarter less than the rectangular conductor,

envelope	$\varnothing 120 \text{ mm} \times 68 \text{ mm}$, 25 mm stack, $\varnothing 64 \text{ mm}$ free bore
slots / poles	24 / 22, slots parallel-sided, 5.8 mm wide, 10 mm deep
coil	one per tooth, a single column of 4 turns
conductor	hollow rectangular copper, $2.4 \times 2.2 \text{ mm}$, $\varnothing 1.2 \text{ mm}$ bore, 4.1 mm^2
no-load air-gap field	1.05 T (working harmonic, peak)
continuous, 850 W total heat	16.6 N m at 132 A rms per phase (32.5 A/mm^2)
heat at that point	copper 793 W, drive and leads 51 W, pump 3 W
coolant	in 60°C , out 71°C , 2.6 L/min, 0.28 bar, $Re \approx 610$
temperatures	skin 61°C , hottest copper 126°C , magnets about 90°C
cooling limit (3 L/min)	21.3 N m at 43 A/mm^2 , about 1.8 kW
torque per amp down 20 %	at 67 A/mm^2 (28.6 N m), beyond the pump
motor mass	about 1.7 kg (estimate, no gear stage)

Table 4: The shallow-slot design, as simulated.

which is a custom draw. Three or five turns per coil in slots 8 mm or 12 mm deep give 15.4 N m and 16.9 N m.

5.4 What the CAD caught

- *Coils sharing a slot collided.* At a conductor pitch of 2.6 mm the two coil sides in one slot were 0.2 mm apart and their end bends crossed as they left the slot. The pitch is now 2.5 mm, 0.4 mm apart, and all thirteen pair checks report no shared volume. The loss and field models were rerun with the new conductor.
- *The end bends are tight.* The bend radius on the conductor centre is 4.2 mm to 5.1 mm for a conductor 2.4 mm wide: under twice its width, on a hollow section. Whether that can be formed without collapsing the bore is a question for a forming trial, and it is the first thing we would test.

5.5 The simpler motor: solid conductors, jacket only

Filling in the bore gives 28 % more copper and removes the manifold, the custom conductor and the oil-to-copper joints. All heat then leaves through the tooth. With coolant at 60°C :

	torque	copper loss	hottest copper
jacket only	10.2 N m	225 W	130°C
jacket only	12.4 N m	385 W	180°C
hollow conductors	16.6 N m	793 W	126°C
hollow conductors, pump limit	21.3 N m	about 1.7 kW	198°C

Table 5: Jacket-only cooling against hollow conductors, same iron and slots.

The jacket alone gives about 75 % of the hollow design’s 850 W torque and 58 % of its limit, with ordinary wire. Figure 2 (right) shows why the comparison is close: per watt of copper loss the two windings give nearly the same torque, and the hollow conductor’s advantage is only that its curve continues. It does this well because each coil is a single column, so every conductor has a face on a tooth wall; the four turns are within 3 % of each other in temperature rise. Of that

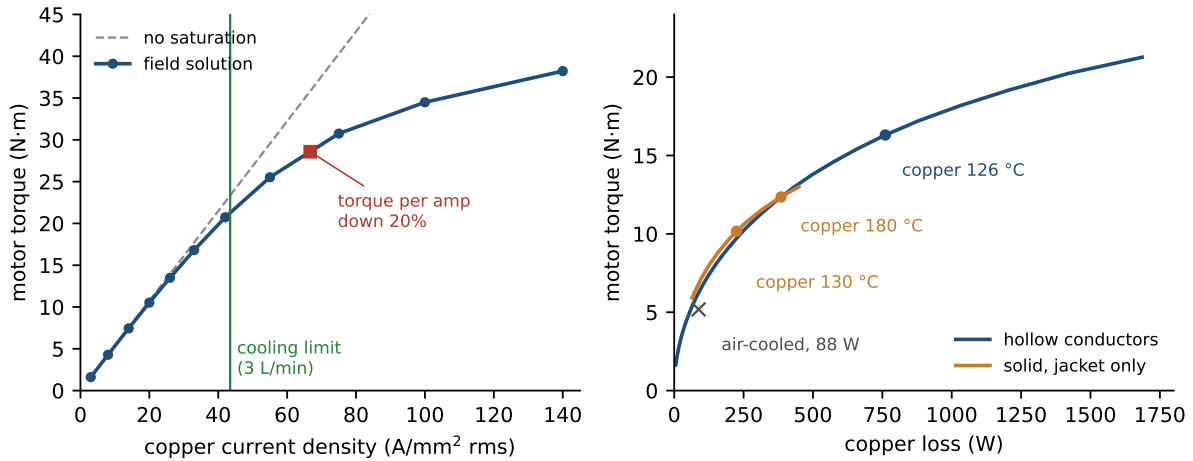


Figure 2: Shallow-slot design. Left: field-solved torque against copper current density. Right: torque against copper loss for hollow conductors and for solid conductors with only the jacket; the curves end where each cooling scheme does. The air-cooled point is the same iron with a conventional winding.

rise, 42 % is across the press fit and coolant film, 20 % in the iron and 38 % across the liner and insulation, so the result depends on how the coil is seated and how the stator is fitted: at 180 °C copper, a loose coil with a 0.05 mm air gap gives 10.6 N m, a poor press fit 10.1 N m, potting at 1 W/m/K 14.9 N m, and potting with a turbulent film and a good fit 18.5 N m, which exceeds the hollow design's 850 W figure.

Our recommendation is therefore to build the jacket-only motor first. It tests the iron, the field solution and the jacket with no unproven manufacturing step, and the hollow conductor becomes an upgrade to the same stator.

5.6 At the joint

ratio	continuous output	heat to hold 35 N m	reflected inertia
9:1	143 N m	43 W	1×
6:1	95 N m	95 W	0.44×
4:1	64 N m	216 W	0.20×
3:1	48 N m	404 W	0.11×
2:1	32 N m	1105 W	0.05×

Table 6: The shallow-slot motor behind a single stage of 96 % efficiency (assumed), at 850 W. Reflected inertia is relative to 9:1 with the same rotor.

The reference actuator gives 35 N m continuously at 9:1, which we estimate costs it about 88 W from its published torque constant and line resistance. At 3:1 the liquid-cooled motor exceeds that continuous torque with a ninth of the reflected inertia and spends about 400 W holding the same load (Table 6). Direct drive is out of reach at this size. The comparison is not like for like in one respect: the reference figure includes its gear stage in 42 mm of length and

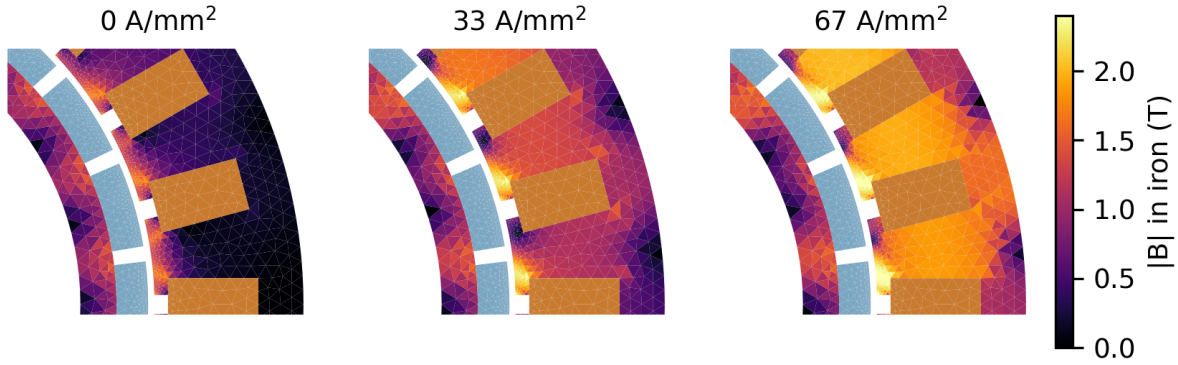


Figure 3: Flux density in the iron of the shallow-slot design at no current, at the 850 W point, and where torque per amp is down 20 %.

1.4 kg; ours is a 68 mm, 1.7 kg motor with no gear stage, a pump, a radiator and hoses still to add.

5.7 Two findings from the sizing model

“Very high current” is a choice of turns. What the copper and the coolant see is current density. Fewer turns raise the phase current and the drive loss without adding torque: at the same torque in the first geometry, 2 turns per coil means about 280 A and 195 W in the switches and leads, and 8 turns means 70 A and 17 W. More turns shrink the bore, which costs pump pressure. A very high phase current is thus a consequence to be minimised, not a goal.

Running the loop hot costs torque and saves radiator. Copper resistance rises 0.39 % per kelvin and the magnets weaken. Raising the coolant return from 60 °C to 100 °C adds about 12 % heat at the same current and needs a higher magnet grade; the return is a radiator a little over half the size.

6 Design workflow

The study was carried out as a loop: define the design in code, simulate it, check the result against an independent model or a geometric test, and change the design when they disagree. Earlier case studies by the author applied this loop to reproduce published designs as verified assemblies.¹ Here it is applied in the other direction, from a hypothesis to a design study.

Roles. The author stated the hypothesis of Section 1, chose the direction at each step (Table 7) and posed the jacket-only comparison. An AI coding agent (Claude, Anthropic, run in Claude Code with shell access to the repository) chose the envelope and topology, wrote the four models and the CAD script, ran them, found and corrected the errors listed below, and drafted the documentation and this paper. Bibliographic details were checked against Crossref records, and cited claims against the sources’ abstracts; the notes are in the repository.

¹For example <https://punkfab.com/blog/vtap-from-paper/>.

	direction given by the author	produced
1	the hypothesis	concept, sizing model, first findings
2	run the solver; is the trade size, for more iron?	nonlinear field solver, eleven-variant study
3	push, then do the shallow slot	shallow-slot design, conductor layouts, CAD assembly and checks
4	what if the conductors were solid, with only a jacket?	jacket-only thermal model and comparison
5	push everything	commits
6	write a blog post	public write-up
7	make some renders	cutaway and stator renders

Table 7: Steps of the study: the direction given at each step and what it produced.

Elapsed time. By the session log, with gaps longer than fifteen minutes excluded, the work in Table 7 took about 70 minutes of agent working time in two sittings on one day (56 minutes for steps 1–3, 13 minutes for steps 4–7). This paper, its figures, the solver checks in Section 4.2 and the thermal field solution were additional. The figure measures elapsed session time, not effort that would compare with a person’s hours, and it excludes the author’s prior work on the hypothesis.

Errors found by cross-checking. The models check each other. In the order found:

1. *A nonsensical rating.* The first sizing run reported a thermal limit at which the motor dissipated 37 kW for no added torque, because the assumed saturation knee was a hard ceiling and the rating looked only at temperature. Caught by reading the output. The rating was redefined, the pump pressure made a least-heat choice, and the flow cap added.
2. *The assumed knee.* The field solver replaced it and showed it pessimistic by a quarter to a third (Table 2).
3. *Two geometry errors.* The outrunner’s slot radius ignored the air gap, and the free bore was computed with the air gap subtracted on the wrong side. Both were found and fixed during the field-solver work.
4. *A silent solver failure.* With parallel-sided slots, the slot ends were tangent to construction circles in the mesh, which produced needle elements; Newton stopped at its iteration cap and returned torque that was too low. Caught because torque per amp was not monotonic in current, which is not physical. The circles were removed from the mesh, the solver now warns when it does not converge, and the affected results were rerun.
5. *The coil clash* and the tight end bends of Section 5, caught by the interference check and by reading the bend radius back from the solids.
6. *An optimistic topology estimate.* The first-order model put an outrunner at 1.5 times the torque; the field solution says 11 %.

Items 2, 4 and 6 changed conclusions. Item 4 is the most serious: a newly written solver returned plausible but wrong values, and only a physical-plausibility check on its output detected them.

Limits of the workflow. Every check above is one model against another, or a model against geometry. None is a measurement. The loop can show that the design is self-consistent and that a conductor fits a slot; it cannot show that a hollow conductor survives its end bend, that a press fit reaches $2000 \text{ W/m}^2/\text{K}$, or that the two-dimensional torque survives a 25 mm stack. Reproducing a published design ends at something that can be compared with the publication; a study that starts from a hypothesis ends at claims that only hardware can test.

7 Limitations

- **No hardware.** Nothing here has been built, and no model has been compared with a measured machine.
- **Two-dimensional field.** A 25 mm stack on a 79 mm gap diameter will lose torque to end leakage that the solver does not see.
- **One rotor position.** No torque ripple or cogging average; no demagnetisation check, although the magnets sit near 90°C beside copper at 126°C and above; iron loss only as a rough estimate; AC copper loss not modelled beyond noting that the skin depth (about 10 mm) far exceeds the conductor.
- **Generic steel.** The B–H curve is a textbook fit; the cobalt-iron variant is that curve stretched by 15 %.
- **Hydraulics.** Bends, manifolds and fittings are not in the pressure drop, and flow is assumed to share equally among 24 parallel coils.
- **Thermal coefficients.** Coolant properties are handbook approximations; the magnet temperature is an interpolation; every coefficient in the jacket-only path is assumed, and Section 5.5 shows the result moving from 10.1 N m to 18.5 N m across plausible values.
- **Manufacture.** A custom hollow conductor, its end bends, its hydraulic and electrical joints, insulation that tolerates oil at 200°C , seals, and the manifold are all undrawn or untested.
- **System.** The pump, radiator, hoses and gear stage are not designed, and their mass is not in any figure here.

8 Conclusion

Within the limits of simulation, the hypothesis mostly holds. Cooling the copper from inside lifts the continuous torque of a $\varnothing 120 \text{ mm}$ joint motor from about 5 N m to between 17 N m and 21 N m and lets the gear ratio fall from 9:1 to about 3:1. Three corrections came out of the work. Heat is not always the only limit: in the first geometry the iron ran out at nearly the current the pump did. The price of a low ratio is energy, since holding torque costs heat in proportion to the square of motor torque. And most of the benefit is available without hollow conductors at all: a single-column coil cooled only through the tooth reaches about 75 % of the torque with none of the manufacturing risk, so that is the motor to build first.

A natural extension is to other actuator types. The jacket-only result suggests that the question to carry forward is less “can coolant reach the copper” than “how short can the path

from every conductor to a cooled wall be made”, which applies equally to linear and axial-flux machines.

Availability

All code, the STEP assembly and the scripts that print every number in this paper are at <https://github.com/punkfab/robot-actuators> under `qdd-liquid/`, MIT-licensed. The `make` targets listed in the README there regenerate the tables and figures; the two FEMM checks need FEMM 4.2. A shorter account is at <https://punkfab.com/blog/liquid-cooled-qdd/>. This preprint is archived at <https://doi.org/10.5281/zenodo.23155222>.

Use of AI

The models, code, figures and text of this paper were produced by an AI coding agent (Claude, Anthropic) directed by the author as described in Section 6. The author set the question and scope, reviewed the result, and is responsible for its content.

References

- [1] Patrick M. Wensing, Albert Wang, Sangok Seok, David Otten, Jeffrey Lang, and Sangbae Kim. Proprioceptive actuator design in the MIT Cheetah: Impact mitigation and high-bandwidth physical interaction for dynamic legged robots. *IEEE Transactions on Robotics*, 33(3):509–522, 2017. doi: 10.1109/TRO.2016.2640183.
- [2] Sangok Seok, Albert Wang, Meng Yee Chuah, Dong Jin Hyun, Jongwoo Lee, David M. Otten, Jeffrey H. Lang, and Sangbae Kim. Design principles for energy-efficient legged locomotion and implementation on the MIT Cheetah robot. *IEEE/ASME Transactions on Mechatronics*, 20(3):1117–1129, 2015. doi: 10.1109/TMECH.2014.2339013.
- [3] Benjamin G. Katz. A low cost modular actuator for dynamic robots. S.M. thesis, Massachusetts Institute of Technology, Department of Mechanical Engineering, 2018. URL <https://dspace.mit.edu/handle/1721.1/118671>.
- [4] Junichi Urata, Yuto Nakanishi, Kei Okada, and Masayuki Inaba. Design of high torque and high speed leg module for high power humanoid. In *2010 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 4497–4502, 2010. doi: 10.1109/IROS.2010.5649683.
- [5] Nicholas Paine and Luis Sentis. Design and comparative analysis of a retrofitted liquid cooling system for high-power actuators. *Actuators*, 4(3):182–202, 2015. doi: 10.3390/act4030182.
- [6] Donghyun Kim, Junhyeok Ahn, Orion Campbell, Nicholas Paine, and Luis Sentis. Investigations of a robotic test bed with viscoelastic liquid cooled actuators. *IEEE/ASME Transactions on Mechatronics*, 23(6):2704–2714, 2018. doi: 10.1109/TMECH.2018.2872176.
- [7] Taoyuanmin Zhu, Joshua Hooks, and Dennis Hong. Design, modeling, and analysis of a liquid cooled proprioceptive actuator for legged robots. In *2019 IEEE/ASME International*

- Conference on Advanced Intelligent Mechatronics (AIM)*, pages 36–43, 2019. doi: 10.1109/AIM.2019.8868596.
- [8] Anirban Mazumdar, Steven J. Spencer, Clinton Hobart, Michael Kuehl, Gregory Brunson, Nadia Coleman, and Stephen P. Buerger. Improving robotic actuator torque density and efficiency through enhanced heat transfer. In *ASME 2016 Dynamic Systems and Control Conference (DSCC), Volume 2*, page V002T26A004, 2016. doi: 10.1115/DSCC2016-9738.
 - [9] Toyotaka Kozuki, Toshinori Hirose, Takuma Shirai, Shinsuke Nakashima, Yuki Asano, Yohei Kakiuchi, Kei Okada, and Masayuki Inaba. Skeletal structure with artificial perspiration for cooling by latent heat for musculoskeletal humanoid Kengoro. In *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 2135–2140, 2016. doi: 10.1109/IROS.2016.7759335.
 - [10] Qingming Li, Haonan Zhou, Deqing Mei, and Yancheng Wang. Highly efficient evaporative cooling of humanoid robot joint motors based on phase change hydrogels. *Applied Thermal Engineering*, 298:131111, 2026. doi: 10.1016/j.applthermaleng.2026.131111.
 - [11] Taoyuanmin Zhu, Min Sung Ahn, and Dennis Hong. Design and experimental study of BLDC motor immersion cooling for legged robots. In *2021 20th International Conference on Advanced Robotics (ICAR)*, pages 1137–1143, 2021. doi: 10.1109/ICAR53236.2021.9659379.
 - [12] Thomas Bauer and Matthias Svoboda. Stator leakage monitoring system in water-cooled generators: Problems and solutions. In *ASME 2021 Power Conference*, 2021. doi: 10.1115/POWER2021-65471.
 - [13] Robert Svoboda, Claudio Picech, and Herbert Hehs. Experiences with stainless steel hollow conductors for generator stator water cooling. In *ASME 2004 Power Conference*, pages 373–377, 2004. doi: 10.1115/POWER2004-52155.
 - [14] Pia M. Lindh, Ilya Petrov, R. Scott Semken, Markku Niemelä, Juha J. Pyrhönen, Lassi Aarniovuori, Toomas Vaimann, and Ants Kallaste. Direct liquid cooling in low-power electrical machines: Proof-of-concept. *IEEE Transactions on Energy Conversion*, 31(4):1257–1266, 2016. doi: 10.1109/TEC.2016.2597059.
 - [15] Pia Lindh, Ilya Petrov, Ahti Jaatinen-Värri, Aki Grönman, Miguel Martínez-Iturralde, Marco Satrustegui, and Juha Pyrhönen. Direct liquid cooling method verified with an axial-flux permanent-magnet traction machine prototype. *IEEE Transactions on Industrial Electronics*, 64(8):6086–6095, 2017. doi: 10.1109/TIE.2017.2681975.
 - [16] Markus Schiefer and Martin Doppelbauer. Indirect slot cooling for high-power-density machines with concentrated winding. In *2015 IEEE International Electric Machines & Drives Conference (IEMDC)*, pages 1820–1825, 2015. doi: 10.1109/IEMDC.2015.7409311.
 - [17] C. Wohlers, P. Juris, S. Kabelac, and B. Ponick. Design and direct liquid cooling of tooth-coil windings. *Electrical Engineering*, 100(4):2299–2308, 2018. doi: 10.1007/s00202-018-0704-x.
 - [18] Fan Wu and Ayman M. EL-Refaie. Additively manufactured hollow conductors with integrated cooling for high specific power electrical machines. In *2020 International Conference*

- on *Electrical Machines (ICEM)*, pages 1497–1503, 2020. doi: 10.1109/ICEM49940.2020.9270871.
- [19] Fan Wu, Ayman EL-Refaie, and Ali Al-Qarni. Additively manufactured hollow conductors integrated with heat pipes: Design tradeoffs and hardware demonstration. *IEEE Transactions on Industry Applications*, 57(4):3632–3642, 2021. doi: 10.1109/TIA.2021.3076423.
 - [20] S. G. J. Geelen, M. Curti, and E. A. Lomonova. Magnetic and thermal modeling of hollow conductors for improved cooling and force density of coreless linear motors. *IEEE Transactions on Magnetics*, 60(9):1–5, 2024. doi: 10.1109/TMAG.2024.3417623.
 - [21] Yaohui Gai, Mohammad Kimiabeigi, Yew Chuan Chong, James D. Widmer, Xu Deng, Mircea Popescu, James Goss, Dave A. Staton, and Andrew Steven. Cooling of automotive traction motors: Schemes, examples, and computation methods. *IEEE Transactions on Industrial Electronics*, 66(3):1681–1692, 2019. doi: 10.1109/TIE.2018.2835397.
 - [22] Yinye Yang, Berker Bilgin, Michael Kasprzak, Shamsuddeen Nalakath, Hossam Sadek, Matthias Preindl, James Cotton, Nigel Schofield, and Ali Emadi. Thermal management of electric machines. *IET Electrical Systems in Transportation*, 7(2):104–116, 2017. doi: 10.1049/iet-est.2015.0050.
 - [23] Juha Pyrhönen, Tapani Jokinen, and Valéria Hrabovcová. *Design of Rotating Electrical Machines*. John Wiley & Sons, Chichester, UK, 2 edition, 2013. ISBN 978-1-118-58157-5. doi: 10.1002/9781118701591.
 - [24] George P. Sutton and Oscar Biblarz. *Rocket Propulsion Elements*. John Wiley & Sons, 9 edition, 2017. ISBN 978-1-118-75365-1.
 - [25] RobStride. RS04 user manual (120 N.M quasi-direct drive integrated motor module) and RS-series specification booklet. https://github.com/RobStride/Product_Information, 2026. Manual revision 260713, booklet dated 2026-09-17. Accessed October 2026.
 - [26] Theodore L. Bergman, Adrienne S. Lavine, Frank P. Incropera, and David P. DeWitt. *Fundamentals of Heat and Mass Transfer*. Wiley, 7 edition, 2011. ISBN 978-0-470-50197-9.
 - [27] Christophe Geuzaine and Jean-François Remacle. Gmsh: A 3-D finite element mesh generator with built-in pre- and post-processing facilities. *International Journal for Numerical Methods in Engineering*, 79(11):1309–1331, 2009. doi: 10.1002/nme.2579.
 - [28] John R. Brauer. Simple equations for the magnetization and reluctivity curves of steel. *IEEE Transactions on Magnetics*, 11(1):81, 1975. doi: 10.1109/TMAG.1975.1058555.
 - [29] Antero Arkkio. *Analysis of Induction Motors Based on the Numerical Solution of the Magnetic Field and Circuit Equations*. PhD thesis, Helsinki University of Technology, Helsinki, Finland, 1987. URL <https://aaltodoc.aalto.fi/handle/123456789/2158>. 97 pp., ISBN 951-22-6076-X.
 - [30] David Meeker. *Finite Element Method Magnetics, Version 4.2, User’s Manual*, January 2018. Software build of 21 April 2019. <https://www.femm.info>.
 - [31] build123d contributors. build123d: A python-based parametric BREP CAD modeling framework. <https://github.com/gumyr/build123d>, 2026. Accessed October 2026.